

a magic cell. It has the ability to self replicate for indefinite periods of time. If you look at haematopoietic stem cells, these are basically blood cells which come from bone marrow. Personalised medicine is nothing but a gene variation. Each one of us has a different genetic code and that is the reason we respond differently to drug treatment. So, drugs given to one individual will respond very differently as compared to another individual. The idea is to understand this genetic variation a bit better.

So, where does India stand in global innovation? India's quest for innovation has been less than 15 years old. India traditionally had the image of being the source of copy-cats of drugs and reverse engineers. However, over the last 15 years Indian companies have taken steps to build innovation as their backbone and clearly Glenmark is at the forefront. Resources for innovation in India are limited. The top five Indian companies have one-fortieth the R&D budget of a Pfizer. Also, there is limited access to external or venture capital funding. If India has to become a global player in pharma innovation, there has to be a succinct effort from both the government and industry to capitalise on the future—mainly in biologics and stem cell research. This is a skillset that India currently does not have which we clearly need to adapt.

"IF INDIA HAS TO BECOME A GLOBAL PLAYER IN PHARMA, IT MUST FOCUS ON STEM CELL RESEARCH."

Glenn Saldanha

DAVID S. SALDANHA



"The access to space is being transformed"

STEPHEN ATTENBOROUGH

COMMERCIAL DIRECTOR, VIRGIN GALACTIC

There is a picture you might be familiar with. Of a blue marble that has become the most reproduced image in history. Everybody knows what earth looks like from space yet no one has seen it with their own eyes. Since Yuri Gagarin went to space for the first time in 1961 over 450 fortunate individuals have followed in his tracks. That's what we are going to change. All the photos that have been taken from space have either been taken by government employees from space agencies around the world or from government satellites. The first time a private citizen, an employee of an aeronautical company in the US, took a picture was from a tiny digital camera belonging to the test pilot of the world's first privately constructed spaceship that went to space three times in 2004.

It was an incredibly significant day for us because it meant that Virgin Galactic was alive. We announced that we would be licensing that technology, developing a commercial vehicle doing exactly what this vehicle did with six passengers on board and two pilots, with a level of safety

similar to aviation so that it can be justifiably called the dawning of a new age. The rocket that you see in front of you was the rocket used to take the Apollo astronauts up. This was the technology that was developed post the Second World War by the United States and the USSR, potentially to shoot a ballistic nuclear missile from one place to another. This was the sort of technology that was adapted to the space race which was also a military race at that time. They didn't think about the cost, they didn't think about the safety of the people. Particularly, they didn't think about the environmental impact, that's for sure.

The children of the 1960s were a generation which grew up on space. John F. Kennedy stood up and said the unthinkable—that America would put man on the moon by the end of the decade. It was impossible but was achieved and the generation that grew up during that time was incredibly inspired by the tremendous breakthroughs. Their parents, I think, probably told them that by 1975 everybody would be going to space and this is a disappointed generation because it expected to follow in the footsteps of those first astronauts. It took three kids from the '60s really to change that. The first was Peter Diamondis, an American who was inspired by Charles Lindbergh, the first American to fly solo across the Atlantic, to put up a \$10 million prize for any company or individual able to build and design a spaceship to fly a man into space. Now, the man who eventually did that is Burt Rutan, one of the, if not the, greatest aviation designer of his gener-